

► Soal Latihan 6.1

3. $f(x) = \frac{1}{2}x$; $x_0 = 2$; $x_1 = 4$; $[2, 4]$

Dapatkan laju perubahan rata-rata ; laju perubahan sesaat untuk f yang diberikan.

misal : $4x = h$ $h = 4 - 2 = 2$

$$f(x_0 + h) = f(2 + 2)$$

$$= f(4)$$

$$f(4) = \frac{1}{8} ; f(2) = \frac{1}{4}$$

Jawab :

• Laju Perubahan Kecepatan Rata-Rata

$$v_r = \frac{f(x_0 + h) - f(x_0)}{h}$$

$$= \frac{f(4) - f(2)}{2}$$

$$= \frac{\frac{1}{8} - \frac{1}{4}}{2}$$

$$= \frac{1 - 2}{16}$$

$$= -\frac{1}{16}$$

• Laju Kecepatan Sesaat

$$\lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

$$\lim_{h \rightarrow 0} \frac{\frac{1}{2(x+h)} - \frac{1}{2x}}{h}$$

$$\lim_{h \rightarrow 0} \frac{1}{h}$$

$$\lim_{h \rightarrow 0} \frac{2x - (2x + 2h)}{4x(x+h)}$$

$$\lim_{h \rightarrow 0} \frac{1}{h}$$

$$\lim_{h \rightarrow 0} \frac{-2}{2x(x+h)}$$

$$= -\frac{1}{2x^2}$$

→ Kecepatan sesaat di $x = 2$

$$v_s = -\frac{1}{2x^2} = -\frac{1}{2(2)^2} = -\frac{1}{8}$$

Jadi, laju perubahan kecepatan rata-ratanya adalah $-\frac{1}{16}$ dan laju kecepatan sesaatnya adalah $-\frac{1}{8}$.

7). $f(x) = x - \sqrt{x}$; $x_0 = 4$

Dapatkan persamaan garis singgung pada kurva $y = f(x)$ di $x = x_0$

► Dengan turunan

$$f'(4) = m = 1 - \frac{1}{2\sqrt{x}}$$

$$= 1 - \frac{1}{4}$$

$$= \frac{3}{4}$$

► Konsep limit mencari gradien

$$\lim_{x \rightarrow 4} \frac{f(x) - f(x_0)}{x - x_0}$$

$$x_0 = 4$$

$$f(x_0) = 4 - \sqrt{4}$$

$$= 2$$

$$= \lim_{x \rightarrow 4} \frac{x - \sqrt{x} - 2}{x - 4}$$

$$= \lim_{x \rightarrow 4} \frac{(\sqrt{x} - 2)(\sqrt{x} + 1)}{(\sqrt{x} - 2)(\sqrt{x} + 2)}$$

$$= \frac{\sqrt{4} + 1}{\sqrt{4} + 2}$$

$$= \frac{3}{4}$$

► Menentukan Persamaan Garis Singgung

$$f(4) = 4 - \sqrt{4}$$

$$= 2$$

• Garis yang bergradien $\frac{3}{4}$ dan melewati $(4; 2)$

$$y - y_1 = m(x - x_1)$$

$$y - 2 = \frac{3}{4}(x - 4)$$

$$y = \frac{3}{4}x - 3 + 2$$

$$y = \frac{3}{4}x - 1$$

Jadi, persamaan garis singgungnya adalah

$$y = \frac{3}{4}x - 1$$

10 Diket: $t_0 = 7$ $t_1 = 11$

$$h = t_1 - t_0$$

$$= 11 - 7$$

$$= 4$$

$$f(t_0) = 15 \quad f(t_1) = 30$$

Ditanya = Cari Informasi tepat

Jawab =

$$b_r = \frac{f(t_0 + h) - f(t_0)}{h}$$

$$= \frac{f(7 + 4) - f(7)}{4}$$

$$= \frac{f(11) - f(7)}{4}$$

$$= \frac{30 - 15}{4}$$

$$= \frac{15}{4}$$

$$= 3,75^\circ \text{C tiap jam}$$

Jadi, kecepatan rata-rata kenaikan temperaturnya adalah $3,75^\circ \text{C} / \text{jam}$.

9. $\rightarrow b(t)$ pada interval $[0, 10]$, $h = 10 - 0 = 10$

$$b_r = \frac{b(t_0 + h) - b(t_0)}{h} = \frac{b(10) - b(0)}{h}$$

$$= \frac{140 - 50}{10}$$

$$= 9 \text{ cm/tahun}$$

$\bullet$ $b(t)$ rata-rata pada interval $[14, 15]$, $h = 15 - 14 = 1$

$$b_r = \frac{b(t_0 + h) - b(t_0)}{h} = \frac{b(15) - b(14)}{1}$$

$$= \frac{170 - 150}{1}$$

$$= 20 \text{ cm/tahun}$$

Jadi, laju pertumbuhan tertinggi setelah 10 tahun adalah pada usia 14 tahun.

17) Gradien garis singgung di $x = x_0$

$$f(x) = x + \sqrt{x} ; x_0 = 1$$

$$\lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(1 + h) - f(1)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(1 + h) + \sqrt{1 + h} - 2}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h + \sqrt{1 + h} - 1}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h}{h} + \lim_{h \rightarrow 0} \frac{\sqrt{1 + h} - 1}{h}$$

$$= 1 + \lim_{h \rightarrow 0} \frac{(\sqrt{1 + h} - 1)(\sqrt{1 + h} + 1)}{h(\sqrt{1 + h} + 1)}$$

$$= 1 + \lim_{h \rightarrow 0} \frac{1 + h - 1}{h(\sqrt{1 + h} + 1)}$$

$$= 1 + \lim_{h \rightarrow 0} \frac{h}{h(\sqrt{1 + h} + 1)}$$

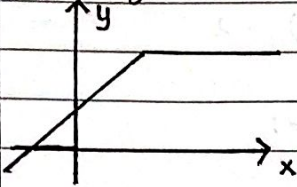
$$= 1 + \frac{1}{2}$$

$$= \frac{3}{2} \quad \text{Jawab: gradien di } x_0 = 1 \text{ adalah } \frac{3}{2}$$

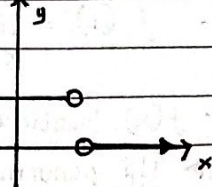
► Soal Latihan 6.2

2. Dapatkan pasangan grafik pada (a) - (d) dengan turunannya pada (A) - (D).

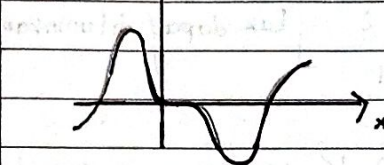
(a). Fungsi :



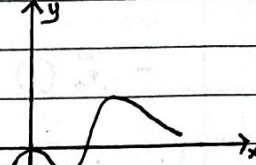
Turunan :



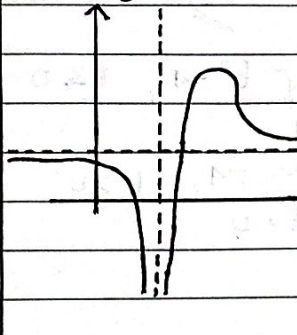
(b). Fungsi :



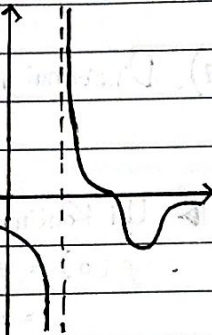
Turunan :



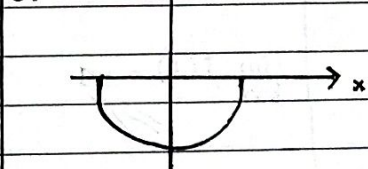
(c). Fungsi :



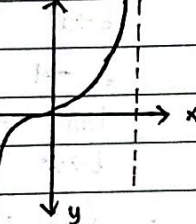
Turunan :



(d). Fungsi :



Turunan :



Jawab :

Pasangan yang tepat adalah (a) - (D) ; (b) - (A) ; (c) - (C) ; dan (d) - (B)

- 8). Dapatkan $f'(x)$ dan persamaan garis yang menyinggung kurva $y = f(x)$ di $x = x_0$

$$f(x) = \sqrt{1-x} ; x_0 = 3/4$$

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x_0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(3/4+h) - f(3/4)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sqrt{1-3/4-h} - \sqrt{1-3/4}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sqrt{1/4-h} - \sqrt{1/4}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sqrt{1/4-h} - 1/2}{h} \cdot \frac{(\sqrt{1/4-h} + 1/2)}{(\sqrt{1/4-h} + 1/2)}$$

$$= \lim_{h \rightarrow 0} \frac{1/4-h - 1/4}{h(\sqrt{1/4-h} + 1/2)}$$

$$= \lim_{h \rightarrow 0} \frac{-h}{h(\sqrt{1/4-h} + 1/2)}$$

$$= \frac{-1}{1}$$

$$= -1$$

• Nilai y

$$\rightarrow y = \sqrt{1-x}$$

$$= \sqrt{1-3/4}$$

$$= \sqrt{1/4}$$

$$= 1/2$$

• Persamaan garis melewati $(3/4, 1/2)$ dan $m = -1$

$$y - y_0 = m(x - x_0)$$

$$y - 1/2 = -1(x - 3/4)$$

$$y = -x + 3/4 + 1/2$$

$$= -x + 5/4$$

$$4y = -4x + 5$$

Jadi, persamaan garisnya menjadi $4y + 4x - 5 = 0$

13). $f(3) = -1$; $f'(3) = 3$

Persamaan garis singgung di kurva $y = f(x)$

pada $x = 3$

$$\text{Gradien} = f'(x) = 3$$

Persamaan garis singgung di $x = 3$ dan $y = -1$
dan memiliki gradien $m = 3$

$$y - y_1 = m(x - x_1)$$

$$y - (-1) = 3(x - 3)$$

$$y + 1 = 3x - 9$$

$$y = 3x - 10$$

Jadi, persamaan garis singgung kurva $y = f(x)$ di
 $x = 3$ adalah $y = 3x - 10$

16. Tunjukkan bahwa :

$$(a). f(x) = \begin{cases} x^2 + 1, & x \leq 1 \\ 2x, & x > 1 \end{cases}$$

Kontinu dan dapat diturunkan di $x = 1$

$$\begin{aligned} \bullet f(1) &= 1^2 + 1 \\ &= 1 + 1 \\ &= 2 \end{aligned}$$

$$\begin{aligned} \bullet \lim_{x \rightarrow 1^-} x^2 + 1 &= 2 \\ \lim_{x \rightarrow 1^+} 2x &= 2 \end{aligned} \quad \left. \begin{array}{l} \lim_{x \rightarrow 1} f(x) = 2 \end{array} \right\}$$

$$\bullet f(1) = \lim_{x \rightarrow 1} f(x) = 2$$

↳ $f(x)$ Kontinu di $x = 1$

$$\begin{aligned} - f'_-(1) &= 2x \\ &= 2 \end{aligned}$$

$$- f'_+(1) = 2$$

Simpulan : $f(x)$ kontinu di $x = 1$ dan memiliki turunan di $x = 1$

$$(b). f(x) = \begin{cases} x^2 + 1, & x \leq 1 \\ x + 1, & x > 1 \end{cases}$$

Kontinu, tetapi tidak dapat diturunkan di $x = 1$

► Uji Kontinuitas

$$\begin{aligned} \bullet f(1) &= 1^2 + 1 \\ &= 2 \end{aligned}$$

$$\begin{aligned} \bullet \lim_{x \rightarrow 1^-} x^2 + 1 &= 2 \\ \lim_{x \rightarrow 1^+} x + 1 &= 2 \end{aligned} \quad \left. \begin{array}{l} \lim_{x \rightarrow 1} f(x) = 2 \end{array} \right\}$$

$$\bullet f(1) = \lim_{x \rightarrow 1} f(x) = 2$$

∴ $f(x)$ Kontinu di $x = 1$

► Uji penurunan

$$\begin{aligned} - f'_-(1) &= 2x \\ &= 2 \\ - f'_+(1) &= 1 \end{aligned} \quad \left. \begin{array}{l} \text{tak dapat diturunkan} \end{array} \right\}$$

Simpulan : $f(x)$ Kontinu di $x = 1$ dan tak dapat diturunkan di $x = 1$

$$17). \text{ Diketahui : } f(t) = \begin{cases} t^2 - 4, & t \leq 0 \\ -4, & t > 0 \end{cases}$$

► Uji Kontinuitas di $t = 0$

$$\begin{aligned} \bullet f(0) &= (0)^2 - 4 \\ &= -4 \end{aligned}$$

$$\begin{aligned} \bullet \lim_{t \rightarrow 0^-} t^2 - 4 &= -4 \\ \lim_{t \rightarrow 0^+} -4 &= -4 \end{aligned} \quad \left. \begin{array}{l} \lim_{t \rightarrow 0} f(t) = -4 \end{array} \right\}$$

$$\bullet f(0) = \lim_{t \rightarrow 0} f(t) = -4$$

Cukup cek di $t = 0$ untuk mengecek $f(t)$ kontinu di mana-mana. Karena kemungkinan $f(t)$ tidak kontinu hanya di $t = 0$. Ketika $f(t)$ kontinu di $t = 0$, maka dapat disimpulkan $f(t)$ kontinu di mana-mana.

► Uji turunan

$$\begin{aligned} - f'_-(0) &= 2t \\ &= 0 \\ - f'_+(0) &= 0 \end{aligned} \quad \left. \begin{array}{l} \text{punya turunan} \end{array} \right\}$$

Simpulan : $f(t)$ Kontinu di mana-mana dan mempunyai turunan di $t = 0$.

► Soal Latihan 6.3

2. $f(x) = 3x^2 - 7x$; $g(x) = 3 - 2x$

$\frac{d}{dx}(f \cdot g)(x) = \dots$

→ $\frac{d}{dx} f(x) \cdot g(x) + f(x) \cdot \left(\frac{d}{dx} g(x) \right)$

$= (6x - 7)(3 - 2x) + (3x^2 - 7x)(-2)$

$= 18x - 12x^2 - 21 + 14x + 14x - 6x^2$

$= -18x^2 + 46x - 21$

→ Pada kunci, seperti soal $g(x)$ seharusnya

$g(x) = 3 - 2x^2$

→ $\left(\frac{d}{dx} f(x) \right) \cdot g(x) + f(x) \left(\frac{d}{dx} g(x) \right)$

$= (6x - 7)(3 - 2x^2) + (3x^2 - 7x)(-4x)$

$= 18x - 12x^2 - 21 + 14x^2 + 28x^2 - 12x^3$

$= -12x^3 + 30x^2 + 18x - 21$

5. $f(-5) = -3$; $f'(-5) = 1/5$. Persamaan garis singgung $y = f(x)$ di $x = -5$

→ Persamaan garis singgung di $(-5, -3)$ dan bergradient $m = 1/5$ adalah ...

$y - y_1 = m(x - x_1)$

$y - (-3) = 1/5(x - (-5))$

$y + 3 = 1/5(x + 5)$

$y = 1/5x + 1 - 3$

$y = 1/5x - 2$

Jadi, persamaan garis singgung $y = f(x)$ adalah

$y = 1/5x - 2$

6. Garis singgung di $x = x_0$ pada kurva $y = \frac{x+3}{x+2}$

tegak lurus garis $y = x$. Dapatkan x_0

$m_1 = -1/m_2$

$m = y' = \frac{(1)(x+2) - (x+3)(1)}{(x+2)^2}$

$= \frac{-1}{(x_0+2)^2} = -1$

$x_0^2 + 4x_0 + 4 = 1$

$x_0^2 + 4x_0 + 3 = 0$

$(x_0 + 3)(x_0 + 1) = 0$

$(x_0 = -3 \vee x_0 = -1)$

Jawab = $x_0 = -3 \vee x_0 = -1$

$\{x | x_0 = -3 \vee x_0 = -1\}$

14). $f'(t)$

a. $f(t) = \frac{2t^2 - t}{t + 1}$

$= \frac{t(2t - 1)}{t(t + 1)}$

$\frac{t(2t - 1)}{t(t + 1)}$

$f'(t) = \frac{2(t+1) - (2t-1)}{(t+1)^2}$

$= \frac{2t + 2 - 2t + 1}{t^2 + 2t + 1}$

$= \frac{3}{t^2 + 2t + 1}$

$= \frac{3}{t^2 + 2t + 1}$

$t^2 + 2t + 1$

b. $f(t) = \frac{(t^3 - 3)(2t - t^2)}{t^4 - t^5 - 6t + 3t^2}$

$= \frac{2t^4 - t^5 - 6t + 3t^2}{t^4 - t^5 - 6t + 3t^2}$

$= -t^5 + 2t^4 + 3t^2 - 6t$

$f'(t) = -5t^4 + 8t^3 + 6t - 6$

15). Persamaan garis singgung pada $y = f(x)$

di $x = -2$; $f(-2) = 3$, $f'(-2) = 1/2$

→ Persamaan garis singgung melalui $(-2, 3)$

dengan gradient $m = 1/2$

$y - y_1 = m(x - x_1)$

$y - 3 = 1/2(x - (-2))$

$2y - 6 = x + 2$

$2y - x - 8 = 0$

Jadi, persamaan garis singgungnya adalah

$2y - x - 8 = 0$

17. $\frac{d^2y}{dx^2}$ dari $y = \frac{(x+1)}{x}$

$$\rightarrow \frac{dy}{dx} = \frac{1(x) - (x+1)}{x^2}$$

$$= \frac{x - x - 1}{x^2}$$

$$= -\frac{1}{x^2}$$

$$\frac{d^2y}{dx^2} = \frac{0(x^2) - (-1)(2x)}{(x^2)^2}$$

$$= \frac{2x}{x^4}$$

$$= \frac{2}{x^3} //$$